

embed a passive, temperature■activated methane■generating chamber upstream of the check valve that releases gas on warm■up to create positive pressure and self■prime the syrup line without electronics

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'loss of prime' concerning syrup pump check valve. User-submitted engineering problem (session input): "Why do syrup-pump check valves in fountain soda machines lose prime and stall after warm restarts, and can a passively self-priming design fix it without electronics?". [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2024-2026). Illustrative records: 'Tracking water vapor homogeneous nucleation and droplet growth with spectroscopy and holography in a free expansion cloud chamber'; 'This report describes a series of tests performed at Sandia's burn site to better understand the behavior of aerosol commodities and their hazards in the transportation environment. This comes on the '. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: methane production by intestinal methanogen overgrowth (increase of methane■producing archaea)

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: embed a passive, temperature■activated methane■generating chamber upstream of the check valve that releases gas on warm■up to create positive pressure and self■prime the syrup line without electronics

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: CONDITIONAL; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: NOT ESTABLISHED.

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected). Engaged variables: dT/dx.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.